TODO

X2 X3 - POWER PINS
vcc+
vcc- - to +/-12v
48v delivery check

0.1u 16v 0402 40
0.1u 25v 0603 60
-1u 60v 0805 2? для 48в пиатльника
-1u 25v 0805 20
4.7uf 25v 1210(B) tantal 16
-10u 5v 0805 22
?10u 25v 0805 - для питаьников? 10
10u 25v 1206() tantal 48
22p 25v 0603 40
220p 25v 0603 12

47u 10v 2917 tantal 4 - для наушников
10n 25v 0603 12

33uH 1280 (shielded) 2

22R 0402 40
510R 0603 64

WM8731
-1n4148 sod 123 8
-led 0805 green 8
red 8

питальники - не подходят пины
PHONE JACKS
PLD приямые и угловые
PBD1.27 smd - для фильтров

-191R 0603 -> 220R 8
-221R

DNP COMPONENTS

FILTER HEADER

CODEC HEADER

INPUT jacks

OUTPUT jacks

SDRAM

MEMORY

FLASH/PSRAM

POWER

Shure SM57 - -56dBV/Pa
60dB SPL -> normal speech
94dB SPL = 1Pa -> max vol

---- my limits -----
120 dB SPL - loudest (mic inside a drum)
20 dB SPL - quetest (ticking watch)

I narrow that range to 60..110dB for quiet mics and 20..80dB SPL for hot mics

4.9V RMS is my ceiling for the +/-12V supply rails

SM78: 60-94= -34dBV (rel to 1Pa) + (-56dBV) = -92dBV = 25uV RMS
120-94=+26 = +10 = 3.2V
<good preamp EIN - 0.1uV 128dBu>
<average preamp EIN - 0.25uV 120dBu>

Hot condenser: 20/94

Mic type	Typical sensitivity	Output at 94 dB SPL	Output at max SPL (close-miked)
Hot condenser (e.g., some LDCs)	−28 dBV/Pa (40 mV/Pa)	40 mV	~1.0 V RMS at 120 dB SPL shout
Typical condenser (SM81)	−45 dBV/Pa (5.6 mV/Pa)	5.6 mV	~0.45 V RMS at 130 dB SPL (snare)
Quiet dynamic (SM78)	−59 dBV/Pa (1.1 mV/Pa)	1.1 mV	~0.09 V RMS at 130 dB SPL
Very quiet passive ribbon	−65 dBV/Pa (0.56 mV/Pa)	0.56 mV	~0.045 V RMS at 130 dB SPL

RE-CHECK PINS
J43_XLR1
NCJ6FA-H
GND_A
+48V_Enabled
R16_XLR1 510 Ohms
C12_XLR1 10 µF
R14_XLR1 5.1 kOhms
R15_XLR1 5.1 kOhms
C1_XLR1 10 µF
R2_XLR1 510 Ohms
R4_XLR1 1 MOhms
100000pF
C7_XLR1 10 µF
R7_XLR1 510 Ohms
IC1B_XLR1
NE5532DRE4
R1_XLR1 510 Ohms
IC1C_XLR1
NE5532DRE4
R5_XLR1 150 kOhms
R9_XLR1 510 Ohms
22 pF C3_XLR1
R8_XLR1 51 kOhms
22 pF C6_XLR1
R11_XLR1 51 kOhms
22 pF
20dB Pad - On SW board
SW2
PushSwitch_DPDT
PAD_SW
R3_XLR1 5.1 kOhms
R13_XLR1 5.1 kOhms
IC2B_XLR1
NE5532DRE4
R6_XLR1 5.1 kOhms
R10_XLR1 5.1 kOhms
22 pF C4_XLR1
XLR_OUT
Audio_testboard[13C]
J1_XLR1
J2_XLR1
SW1
Slide Switch_DPDT
TRS_TIP
TRS_RING
Line_IN
Mic/Line
XLR_A
XLR_B
C14_XLR1 10 µF
VCC+
R17_XLR1 51 kOhms
IC2C_XLR1
NE5532DRE4
R19_XLR1 51 kOhms
Clipping indicator
R18_XLR1 510 Ohms
D1_XLR1
1N4148W-TP
R20_XLR1 5.1 kOhms
C13_XLR1 1 µF
D2_XLR1
Red
R63_XLR1 510 Ohms
C96_XLR1 1 µF
R64_XLR1 5.1 kOhms
D4_XLR1
Green
Q3_XLR1
MMBT3904-7-F
R65_XLR1 5.1 kOhms
R66_XLR1 510 Ohms
X1_XLR1
+48V_Enabled
TRS_RING
Line_IN
TRS_TIP
PAD_SW
XLR_A
GND_A
XLR_B
C8_XLR1 0.1 µF
C10_XLR1 0.1 µF
IC1A_XLR1
NE5532DRE4
C9_XLR1 0.1 µF
C11_XLR1 0.1 µF
IC2A_XLR1
NE5532DRE4
Audio_testboard[12C]
48V
MH
MH5_XLR1

Title		
Size: A3	Number	Revision
Date: 10/20/2026		Sheet of
File: D:\TMP\..\XLR.SchDoc		Drawn By:

Shure SM57 - -56dBV/Pa
60dB SPL -> normal speech
94dB SPL = 1Pa -> max vol

---- my limits -----
120 dB SPL - loudest (mic inside a drum)
20 dB SPL - quetest (ticking watch)

I narrow that range to 60..110dB for quiet mics and 20..80dB SPL for hot mics

4.9V RMS is my ceiling for the +/-12V supply rails

SM78: 60-94= -34dBV (rel to 1Pa) + (-56dBV) = -92dBV = 25uV RMS
120-94=+26 = +10 = 3.2V
<good preamp EIN - 0.1uV 128dBu>
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Mic type	Typical sensitivity	Output at 94 dB SPL	Output at max SPL (close-miked)
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Very quiet passive ribbon	−65 dBV/Pa (0.56 mV/Pa)	0.56 mV	~0.045 V RMS at 130 dB SPL

RE-CHECK PINS
J43_XLR2
NCJ6FA-H
GND_A
+48V_Enabled
R16_XLR2 510 Ohms
C12_XLR2 10 µF
R14_XLR2 5.1 kOhms
R15_XLR2 5.1 kOhms
C1_XLR2 10 µF
R2_XLR2 510 Ohms
IC1B_XLR2
NE5532DRE4
R1_XLR2 MT 510 Ohms
IC1C_XLR2
NE5532DRE4
20dB Pad - On SW board
SW2
PushSwitch_DPDT
PAD_SW
R3_XLR2 5.1 kOhms
R13_XLR2 5.1 kOhms
IC2B_XLR2
NE5532DRE4
XLR_OUT
Audio_testboard[13C]
J1_XLR2
R6_XLR2 5.1 kOhms
R10_XLR2 5.1 kOhms
22 pF C4_XLR2
R4_XLR2 1 MOhms
100000 pF C3_XLR2
C7_XLR2 10 µF
R7_XLR2 510 Ohms
R8_XLR2 51 kOhms
22 pF C6_XLR2
R11_XLR2 51 kOhms
22 pF C5_XLR2
R9_XLR2 510 Ohms
R5_XLR2 150 kOhms
J2_XLR2
SW1
TRS_TIP
TRS_RING
Line_IN
Mic/Line
XLR_A
XLR_B
C14_XLR2 10 µF
VCC+
R17_XLR2 51 kOhms
R19_XLR2 51 kOhms
IC2C_XLR2
NE5532DRE4
Clipping indicator
R18_XLR2 510 Ohms
D1_XLR2 1N4148W-TP
C13_XLR2 1 µF
R20_XLR2 5.1 kOhms
D2_XLR2 Red
R63_XLR2 510 Ohms
C96_XLR2 1 µF
R64_XLR2 5.1 kOhms
D4_XLR2 Green
Q3_XLR2 MMBT3904
R65_XLR2 5.1 kOhms
R66_XLR2 510 Ohms
X1_XLR2
+48V_Enabled
TRS_RING
Line_IN
TRS_TIP
PAD_SW
XLR_A
GND_A
XLR_B
C8_XLR2 0.1 µF
C10_XLR2 0.1 µF
IC1A_XLR2
NE5532DRE4
C9_XLR2 0.1 µF
C11_XLR2 0.1 µF
IC2A_XLR2
NE5532DRE4
Audio_testboard[12C]
48V
MH
MH5_XLR2

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4.9V RMS is my ceiling for the +/-12V supply rails

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120-94=+26 = +10 = 3.2V
<good preamp EIN - 0.1uV 128dBu>
<average preamp EIN - 0.25uV 120dBu>

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Very quiet passive ribbon	−65 dBV/Pa (0.56 mV/Pa)	0.56 mV	~0.045 V RMS at 130 dB SPL

RE-CHECK PINS
J43_XLR3
NCJ6FA-H
GND_A
+48V_Enabled
R16_XLR3 510 Ohms
C12_XLR3 10 µF
R14_XLR3 5.1 kOhms
R15_XLR3 5.1 kOhms
C1_XLR3 10 µF
R2_XLR3 510 Ohms
IC1B_XLR3
NE5532DRE4
R1_XLR3 MT 510 Ohms
IC1C_XLR3
NE5532DRE4
R3_XLR3 5.1 kOhms
R13_XLR3 5.1 kOhms
20dB Pad - On SW board
SW2
PushSwitch_DPDT
PAD_SW
IC2B_XLR3
NE5532DRE4
XLR_OUT
Audio_testboard[13C]
J1_XLR3
R6_XLR3 5.1 kOhms
R10_XLR3 5.1 kOhms
22 pFC4_XLR3
R4_XLR3 1 MOhms
100000pF
C7_XLR3 10 µF
R7_XLR3 510 Ohms
R8_XLR3 51 kOhms
22 pFC6_XLR3
R11_XLR3 51 kOhms
22 pF
R5_XLR3 150 kOhms
R9_XLR3 510 Ohms
22 pFC3_XLR3
J2_XLR3
XLR_A
XLR_B
SW1
TRS_TIP
TRS_RING
Line_IN
Mic/Line
SlideSwitch_DPDT
C14_XLR3 10 µF
VCC+
R17_XLR3 51 kOhms
IC2C_XLR3
NE5532DRE4
R19_XLR3 51 kOhms
Clipping indicator
R18_XLR3 510 Ohms
D1_XLR3
1N4148W-TP
R20_XLR3 5.1 kOhms
C13_XLR3 1 µF
D2_XLR3
Red
R63_XLR3 510 Ohms
C96_XLR3 1 µF
R64_XLR3 5.1 kOhms
D4_XLR3
Green
Q3_XLR3
MMBT3904-7-F
R65_XLR3 5.1 kOhms
R66_XLR3 510 Ohms
X1_XLR3
+48V_Enabled
TRS_RING
Line_IN
TRS_TIP
PAD_SW
XLR_A
GND_A
XLR_B
C8_XLR3 0.1 µF
C10_XLR3 0.1 µF
IC1A_XLR3
NE5532DRE4
C9_XLR3 0.1 µF
C11_XLR3 0.1 µF
IC2A_XLR3
NE5532DRE4
Audio_testboard[12C]
48V
MH
MH5_XLR3

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4.9V RMS is my ceiling for the +/-12V supply rails

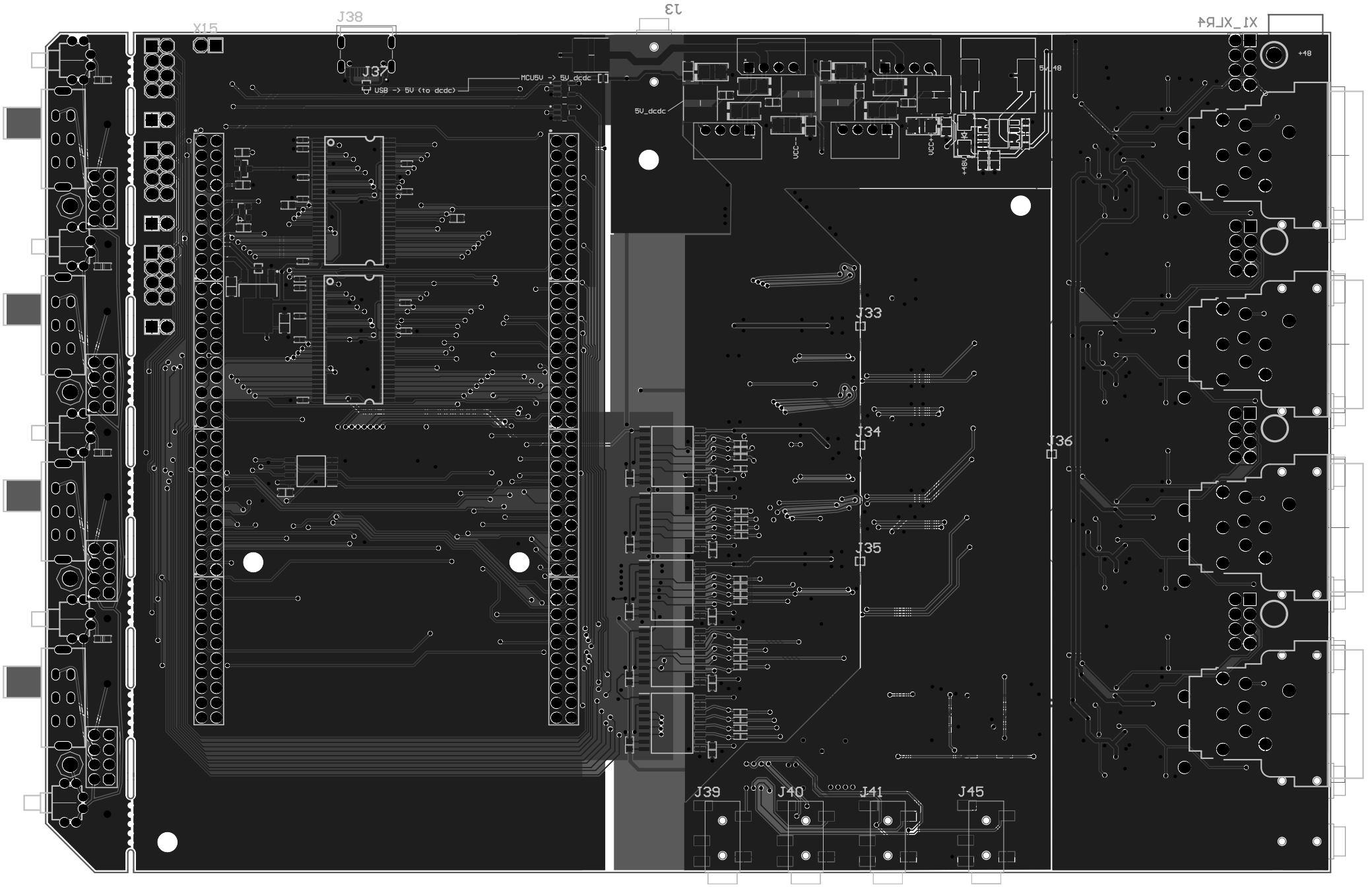
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RE-CHECK PINS
J43_XLR4
NCJ6FA-H
GND_A
+48V_Enabled
R16_XLR4 510 Ohms
C12_XLR4 10 µF
R14_XLR4 5.1 kOhms
R15_XLR4 5.1 kOhms
C1_XLR4 10 µF
R2_XLR4 510 Ohms
IC1B_XLR4
NE5532DRE4
R1_XLR4 MT 510 Ohms
IC1C_XLR4
NE5532DRE4
R3_XLR4 5.1 kOhms
R13_XLR4 5.1 kOhms
20dB Pad - On SW board
SW2
PushSwitch_DPDT
PAD_SW
IC2B_XLR4
NE5532DRE4
XLR_OUT
Audio_testboard[13C]
J1_XLR4
R6_XLR4 5.1 kOhms
R10_XLR4 5.1 kOhms
22 pFC4_XLR4
R4_XLR4 1 MOhms
100000pF
C7_XLR4 10 µF
R7_XLR4 510 Ohms
R8_XLR4 51 kOhms
22 pFC6_XLR4
R11_XLR4 51 kOhms
22 pF
R5_XLR4 150 kOhms
R9_XLR4 510 Ohms
22 pFC3_XLR4
J2_XLR4
XLR_A
XLR_B
SW1
TRS_TIP
TRS_RING
Line_IN
Mic/Line
C14_XLR4 10 µF
VCC+
R17_XLR4 51 kOhms
IC2C_XLR4
NE5532DRE4
R19_XLR4 51 kOhms
Clipping indicator
R18_XLR4 510 Ohms
D1_XLR4
1N4148W-TP
R20_XLR4 5.1 kOhms
C13_XLR4 1 µF
D2_XLR4
Red
R63_XLR4 510 Ohms
C96_XLR4 1 µF
R64_XLR4 5.1 kOhms
D4_XLR4
Green
Q3_XLR4
MMBT3904-7-F
R65_XLR4 5.1 kOhms
R66_XLR4 510 Ohms
X1_XLR4
+48V_Enabled
TRS_RING
Line_IN
TRS_TIP
PAD_SW
XLR_A
GND_A
XLR_B
C8_XLR4 0.1 µF
C10_XLR4 0.1 µF
IC1A_XLR4
NE5532DRE4
C9_XLR4 0.1 µF
C11_XLR4 0.1 µF
IC2A_XLR4
NE5532DRE4
Audio_testboard[12C]
48V
MH
MH5_XLR4

Title		
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J38
X15
J37
MCU5V -> 5V_dcdc
USB -> 5V (to dcdc)
5V_dcdc
J33
J34
J36
J35
J39
J40
J41
J45